Aspiring Business Analyst

Project Portfolio

Analysis & Solving Related to Real-Time Business Problem based on a Five- different project .

Twinkle Rai

Project 1 — E-commerce Drop-off Analysis

1st SQL query -
SESSION - LEVEL FUNNEL ANALYSIS

```sql
/*  Build the Session-Level Funnel
Step 1 - Create session funnel table */
USE ecommerce_dropoff;
GO

SELECT
    user_session,

    MAX(CASE
        WHEN event_type = 'view' THEN 1
        ELSE 0
    END) AS Has_View,

    MAX(CASE
        WHEN event_type = 'cart' THEN 1
        ELSE 0
    END) AS Has_Cart,

    MAX(CASE
        WHEN event_type = 'purchase' THEN 1
        ELSE 0
    END) AS Has_Purchase

INTO dbo.session_funnel
FROM dbo.ecommerce_events
GROUP BY user_session;

SELECT TOP 20 *
FROM [dbo].[session_funnel];
```

user_session	Has_View	Has_Cart	Has_Purchase
0fd72f2c-8e6b-4e9d-a318-398bc909d443	1	0	0
0254e131-2043-4f36-afee-84c3d711dcc5	1	0	0
01489ca4-e14a-466c-b978-6dd820402bc3	1	0	0
09236e51-47ea-4ac0-a8e8-9407d56ad124	1	0	0
22b36a8d-0b12-4859-a2a6-edc549c4845f	1	0	0
2dde26b5-9c02-4a6b-9bb5-d663d67f9e07	1	0	0
f27b75e5-71a0-4b72-8790-ff0113af418a	1	0	0
f28ea226-1dfb-4b07-9f21-923a5f4a74a7	1	0	0
fdbedcc3-1919-467e-bcdf-36963f7d6879	1	0	0
a914af18-aa72-4063-b81d-532b7ed8413e	1	0	0
b2be0c0c-fc53-455e-a083-05386242e96c	1	0	0
b6aa2609-f16d-48d4-8c99-fdf70afbd357	1	0	0
bdacd48a-5c8a-4b97-9fab-2296b8c311fe	1	0	0
c22cb6ff-5a30-4935-a88e-662157d5359b	1	0	0
ed5921b4-5fb9-467c-969a-5e95e05d0486	1	0	0
5e63d21f-27af-4f1f-845e-2c4331a219ea	1	0	0
543086d7-7c3a-48a3-83f3-3c99ca9975e8	1	0	0
60864190-329d-4db5-8be8-677b295b7a1f	1	0	0
71d953a1-4c33-48ed-aca1-d587c0b90dfc	1	0	0
7bb495d0-c602-46f5-81fe-785bca340d5c	1	0	0

Project 1 — E-commerce Drop-off Analysis

1st Step -
COUNT SESSIONS AT EACH FUNNEL STAGE

```sql
SELECT
    COUNT(*) AS Total_Sessions,

    SUM(Has_View) AS View_Sessions,

    SUM(CASE
        WHEN Has_View = 1 AND Has_Cart = 1
        THEN 1 ELSE 0
    END) AS Cart_Sessions,

    SUM(CASE
        WHEN Has_View = 1 AND Has_Cart = 1 AND Has_Purchase = 1
        THEN 1 ELSE 0
    END) AS Purchase_Sessions

FROM [dbo].[session_funnel] ;
```

	Total_Sessions	View_Sessions	Cart_Sessions	Purchase_Sessions
1	117455	117406	5066	2875

Project 1 — E-commerce Drop-off Analysis

2 Step -
CALCULATE FUNNEL CONVERSION & DROP-OFF %

```sql
SELECT
    View_Sessions,
    Cart_Sessions,
    Purchase_Sessions,

    CAST(Cart_Sessions * 100.0 / NULLIF(View_Sessions, 0) AS DECIMAL(10,2))
        AS View_to_Cart_Conversion,

    CAST(Purchase_Sessions * 100.0 / NULLIF(Cart_Sessions, 0) AS DECIMAL(10,2))
        AS Cart_to_Purchase_Conversion,

    CAST(Purchase_Sessions * 100.0 / NULLIF(View_Sessions, 0) AS DECIMAL(10,2))
        AS View_to_Purchase_Conversion,

    CAST(100 - (Cart_Sessions * 100.0 / NULLIF(View_Sessions, 0)) AS DECIMAL(10,2))
        AS View_to_Cart_Dropoff,

    CAST(100 - (Purchase_Sessions * 100.0 / NULLIF(Cart_Sessions, 0)) AS DECIMAL(10,2))
        AS Cart_to_Purchase_Dropoff,

    CAST(100 - (Purchase_Sessions * 100.0 / NULLIF(View_Sessions, 0)) AS DECIMAL(10,2))
        AS Overall_Dropoff

FROM
(
    SELECT
        SUM(Has_View) AS View_Sessions,

        SUM(CASE
            WHEN Has_View = 1 AND Has_Cart = 1
            THEN 1 ELSE 0
        END) AS Cart_Sessions,

        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) AS Purchase_Sessions

    FROM [dbo].[session_funnel]
) AS Funnel;
```

View_Sessions	Cart_Sessions	Purchase_Sessions	View_to_Cart_Conversion	Cart_to_Purchase_Conversion	View_to_Purchase_Conversion	View_to_Cart_Dropoff	Cart_to_Purchase_Dropoff	Overall_Dropoff
117406	5066	2875	4.31	56.75	2.45	95.69	43.25	97.55

Project 1 — E-commerce Drop-off Analysis

2 SQL Query-
CATEGORY - LEVEL FUNNEL SESSION

```sql
USE ecommerce_dropoff;
GO

DROP TABLE IF EXISTS dbo.category_session_funnel;
GO

SELECT
    category_code,
    user_session,

    MAX(CASE
        WHEN event_type = 'view' THEN 1
        ELSE 0
    END) AS Has_View,

    MAX(CASE
        WHEN event_type = 'cart' THEN 1
        ELSE 0
    END) AS Has_Cart,

    MAX(CASE
        WHEN event_type = 'purchase' THEN 1
        ELSE 0
    END) AS Has_Purchase

INTO dbo.category_session_funnel
FROM dbo.ecommerce_events

WHERE category_code IS NOT NULL

GROUP BY
    category_code,
    user_session;
GO

SELECT TOP 20 *              --- step 2
FROM [dbo].[category_session_funnel];
```

	category_code	user_session	Has_View	Has_Cart	Has_Purchase
1	computers.peripherals.keyboard	0002c5ea-3509-4d0b-9618-7e40925005f0	1	0	1
2	electronics.smartphone	000836b3-154e-46eb-8da5-9a486dc1d251	1	0	0
3	electronics.smartphone	000c2fd2-561a-4abd-bd21-5e71449b1b78	1	0	0
4	electronics.smartphone	0017074c-a5fd-4422-80c8-5498a1e75cf8	1	0	0
5	electronics.audio.subwoofer	001dbb69-e6ca-47eb-87e3-5dc4ccdb9199	1	0	0
6	electronics.smartphone	00242532-a09f-4159-bc7b-c1cd699dc990	1	0	0
7	electronics.smartphone	002739d4-9097-4c71-a871-98a83a645ce6	1	0	0
8	electronics.smartphone	0027e42c-d3cc-4976-a3fb-0aa4c1906d30	1	0	0
9	electronics.smartphone	002c8f5c-3fcb-4d0e-82f8-a5f70e66317f	1	0	0
10	electronics.smartphone	0032c643-5d50-4b9f-b97b-6582f7b3f236	1	0	0
11	electronics.smartphone	00399d10-e0d3-4d17-9760-30c42851777c	1	0	0
12	appliances.environment.vacuum	003b1ba8-0852-4a15-a2b2-42470979f085	1	0	0
13	electronics.smartphone	004adc3c-1e2b-45bd-928a-15bebc9e0b9f	1	0	0
14	electronics.smartphone	004c7ccf-f582-4cae-894e-35129313fa45	1	0	0
15	appliances.environment.air_he...	0050a463-f560-42e5-b1d3-76c8c7b57ff3	1	0	0
16	electronics.tablet	005cc7e4-9de4-4eee-a20d-c33653182d77	1	0	0
17	electronics.smartphone	0062b869-3009-4c86-b005-8efcd725c578	1	0	0
18	electronics.tablet	0070b886-5f23-4296-99c4-9bdd368faab4	1	0	0
19	apparel.shoes	00712e9b-2be2-49ee-92f6-b2d4cebc2754	1	0	0
20	appliances.kitchen.oven	0072e561-56c4-435c-9a8c-4a62f14f848e	1	0	0

Project 1 — E-commerce Drop-off Analysis

1st step-
CATEGORY LEVEL FUNNEL

```sql
SELECT
    category_code,

    SUM(Has_View) AS View_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
        THEN 1
        ELSE 0
    END) AS Cart_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
         AND Has_Purchase = 1
        THEN 1
        ELSE 0
    END) AS Purchase_Sessions

FROM dbo.category_session_funnel

GROUP BY category_code

ORDER BY View_Sessions DESC;
```

	category_code	View_Sessions	Cart_Sessions	Purchase_Sessions
1	electronics.smartphone	39204	3606	2232
2	electronics.clocks	4943	130	70
3	electronics.audio.headphone	4737	323	154
4	computers.notebook	3302	21	12
5	electronics.video.tv	2784	154	76
6	appliances.kitchen.refrigerators	2489	9	4
7	appliances.environment.vacuum	2251	57	31
8	apparel.shoes	1994	0	0
9	appliances.kitchen.washer	1917	33	16
10	auto.accessories.player	1608	5	0
11	apparel.shoes.keds	1578	0	0
12	computers.desktop	1263	2	0
13	furniture.bedroom.bed	1203	1	1
14	electronics.tablet	1049	26	14
15	appliances.environment.air_he...	1044	12	3
16	electronics.audio.subwoofer	1011	1	1
17	electronics.telephone	973	32	13

Project 1 — E-commerce Drop-off Analysis

2 step-
CATEGORY CONVERSION & DROP-OFF ANALYSIS

```sql
SELECT
    category_code,

    SUM(Has_View) AS View_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
        THEN 1
        ELSE 0
    END) AS Cart_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
         AND Has_Purchase = 1
        THEN 1
        ELSE 0
    END) AS Purchase_Sessions,

    ROUND(
        100.0 *
        SUM(CASE
            WHEN Has_View = 1 AND Has_Cart = 1
            THEN 1 ELSE 0
        END)
        / NULLIF(SUM(Has_View), 0),
        2
    ) AS View_to_Cart_Conversion,

    ROUND(
        100.0 *
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END)
        / NULLIF(
            SUM(CASE
                WHEN Has_View = 1 AND Has_Cart = 1
                THEN 1 ELSE 0
            END), 0
        ),
        2
    ) AS Cart_to_Purchase_Conversion,

    ROUND(
        100.0 *
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END)
        / NULLIF(SUM(Has_View), 0),
        2
    ) AS View_to_Purchase_Conversion

FROM [dbo].[category_session_funnel]

GROUP BY category_code

HAVING SUM(Has_View) >= 500

ORDER BY View_to_Purchase_Conversion DESC;
```

	category_code	View_Sessions	Cart_Sessions	Purchase_Sessions	View_to_Cart_Conversion	Cart_to_Purchase_Conversion	View_to_Purchase_Conversion
1	electronics.smartphone	39204	3606	2232	9.200000000000	61.900000000000	5.690000000000
2	electronics.audio.headphone	4737	323	154	6.820000000000	47.680000000000	3.250000000000
3	electronics.video.tv	2784	154	76	5.530000000000	49.350000000000	2.730000000000
4	appliances.environment.water_heater	596	22	15	3.690000000000	68.180000000000	2.520000000000
5	construction.tools.saw	665	19	11	2.860000000000	57.890000000000	1.650000000000
6	electronics.clocks	4943	130	70	2.630000000000	53.850000000000	1.420000000000
7	appliances.environment.vacuum	2251	57	31	2.530000000000	54.390000000000	1.380000000000
8	electronics.telephone	973	32	13	3.290000000000	40.630000000000	1.340000000000
9	electronics.tablet	1049	26	14	2.480000000000	53.850000000000	1.330000000000
10	computers.peripherals.printer	573	18	7	3.140000000000	38.890000000000	1.220000000000
11	appliances.kitchen.blender	680	10	8	1.470000000000	80.000000000000	1.180000000000
12	construction.tools.drill	863	16	10	1.850000000000	62.500000000000	1.160000000000
13	auto.accessories.alarm	699	28	7	4.010000000000	25.000000000000	1.000000000000
14	auto.accessories.videoregister	794	15	7	1.890000000000	46.670000000000	0.880000000000
15	appliances.kitchen.washer	1917	33	16	1.720000000000	48.480000000000	0.830000000000
16	appliances.kitchen.microwave	541	5	4	0.920000000000	80.000000000000	0.740000000000
17	appliances.kitchen.kettle	672	5	4	0.740000000000	80.000000000000	0.600000000000

Project 1 — E-commerce Drop-off Analysis

3rd SQL Query -
BRAND LEVEL SESSION FUNNEL

```sql
INSERT INTO dbo.brand_session_funnel
(
    brand,
    user_session,
    Has_View,
    Has_Cart,
    Has_Purchase
)

SELECT
    brand,
    user_session,

    MAX(CASE
        WHEN event_type = 'view' THEN 1
        ELSE 0
    END) AS Has_View,

    MAX(CASE
        WHEN event_type = 'cart' THEN 1
        ELSE 0
    END) AS Has_Cart,

    MAX(CASE
        WHEN event_type = 'purchase' THEN 1
        ELSE 0
    END) AS Has_Purchase

FROM dbo.ecommerce_events

WHERE brand IS NOT NULL

GROUP BY
    brand,
    user_session;

SELECT TOP 20 *             --- checking that
FROM dbo.brand_session_funnel;
```

	brand	user_session	Has_View	Has_Cart	Has_Purchase
1	x-game	0002c5ea-3509-4d0b-9618-7e40925005f0	1	0	1
2	dausc...	00054715-a7ed-42d6-b891-98cd15d7e7...	1	0	0
3	bertoni	000fa659-9b22-4f95-a06b-52cd45382fe3	1	0	0
4	toyota	00101be1-45f5-42da-b48e-cef175e58cb4	1	0	0
5	jaguar	0018a117-207f-4086-a3a4-c14c8f4ca347	1	0	0
6	sams...	0018a117-207f-4086-a3a4-c14c8f4ca347	1	0	0
7	nitro	0019d67a-bfd0-408c-9554-166419bf100a	1	0	0
8	shivaki	0020f3c3-71cd-4d8f-8a1e-e559f7979a4d	1	0	0
9	electr...	00216f71-ed96-47b3-93ec-124530b13a07	1	0	0
10	lucente	002170fe-a352-4f75-b50c-88daa948f41d	1	0	0
11	apple	002c8f5c-3fcb-4d0e-82f8-a5f70e66317f	1	0	0
12	cantra	0030f28a-f77f-4f04-b225-4687f70b96ec	1	0	0
13	sams...	0032c643-5d50-4b9f-b97b-6582f7b3f236	1	0	0
14	apple	00399d10-e0d3-4d17-9760-30c4285177...	1	0	0
15	xiaomi	003b1ba8-0852-4a15-a2b2-42470979f085	1	0	0
16	vitek	003bbed9-f514-4791-9c4e-afd5c57c75a3	1	0	0
17	xiaomi	003f4d74-bbc6-43a1-9e99-18c9cbcf486f	1	0	0
18	lucente	00458e73-ea8d-474a-a175-35a7ac1126...	1	0	0
19	asus	004a1ae0-f2c3-405d-adb7-6d808d97ffbf	1	0	0
20	apple	004c7ccf-f582-4cae-894e-35129313fa45	1	0	0

Project 1 — E-commerce Drop-off Analysis

1st Step-
BRAND LEVEL CONVERSION ANALYSIS

```sql
SELECT
    brand,

    SUM(Has_View) AS View_Sessions,

    SUM(CASE
        WHEN Has_View = 1 AND Has_Cart = 1
        THEN 1 ELSE 0
    END) AS Cart_Sessions,

    SUM(CASE
        WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
        THEN 1 ELSE 0
    END) AS Purchase_Sessions,

    CAST(
        SUM(CASE
            WHEN Has_View = 1 AND Has_Cart = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(SUM(Has_View), 0)
        AS DECIMAL(10,2)
    ) AS View_to_Cart_Conversion,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
                 AND Has_Cart = 1
                 AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(
            SUM(CASE
                WHEN Has_View = 1 AND Has_Cart = 1
                THEN 1 ELSE 0
            END), 0
        )
        AS DECIMAL(10,2)
    ) AS Cart_to_Purchase_Conversion,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
                 AND Has_Cart = 1
                 AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(SUM(Has_View), 0)
        AS DECIMAL(10,2)
    ) AS View_to_Purchase_Conversion

FROM [dbo].[brand_session_funnel]

GROUP BY brand

HAVING SUM(Has_View) >= 500

ORDER BY View_to_Purchase_Conversion DESC;
```

	brand	View_Sessions	Cart_Sessions	Purchase_Sessions	View_to_Cart_Conversion	Cart_to_Purchase_Conversion	View_to_Purchase_Conversion
1	samsung	20934	1884	1198	9.00	63.59	5.72
2	apple	19364	1493	894	7.71	59.88	4.62
3	oppo	1655	114	72	6.89	63.16	4.35
4	huawei	5011	286	176	5.71	61.54	3.51
5	xiaomi	11627	683	298	5.87	43.63	2.56
6	nokian	840	33	11	3.93	33.33	1.31
7	starline	535	28	7	5.23	25.00	1.31
8	redmond	1362	24	15	1.76	62.50	1.10
9	lg	2115	36	23	1.70	63.89	1.09
10	nokia	865	19	9	2.20	47.37	1.04
11	sony	1919	41	19	2.14	46.34	0.99
12	artel	1659	30	15	1.81	50.00	0.90
13	jbl	703	15	5	2.13	33.33	0.71
14	meizu	787	11	5	1.40	45.45	0.64
15	cordiant	1725	48	11	2.78	22.92	0.64
16	vivo	838	6	5	0.72	83.33	0.60

Project 1 — E-commerce Drop-off Analysis

4th SQL Query-
PRICE BAND FUNNEL ANALYSIS

```sql
INSERT INTO [dbo].[price_session_funnel]
(
    price_band,
    user_session,
    Has_View,
    Has_Cart,
    Has_Purchase
)
SELECT
    price_band,
    user_session,

    MAX(CASE
        WHEN event_type = 'view' THEN 1
        ELSE 0
    END) AS Has_View,

    MAX(CASE
        WHEN event_type = 'cart' THEN 1
        ELSE 0
    END) AS Has_Cart,

    MAX(CASE
        WHEN event_type = 'purchase' THEN 1
        ELSE 0
    END) AS Has_Purchase

FROM [dbo].[ecommerce_events]
WHERE price_band IS NOT NULL
GROUP BY
    price_band,
    user_session;

SELECT TOP 20 *           ---just checking f
FROM dbo.price_session_funnel;
```

	price_band	user_session	Has_View	Has_Cart	Has_Purchase
1	100-250	04866be0-00bc-4fed-92fa-4abc716aed77	1	0	0
2	100-250	19010980-dc1e-48dd-99e4-5ee1f8f4790a	1	0	0
3	100-250	23f5bd18-9389-4ee1-8fc0-a1aa03106e42	1	0	0
4	500-1000	29729647-0e2c-23e9-97ff-28d771fbf5bf	1	0	0
5	250-500	3b112286-9563-4541-9d09-5127bd986...	1	0	0
6	1000+	3bc5fc48-10ee-4da2-a321-ad336a6c33...	1	0	0
7	500-1000	b82db56f-8d2f-4e6a-94e2-ecb7e202a7b5	1	0	0
8	50-100	cceaf85a-0319-4cbc-866d-8097902623ae	1	0	0
9	1000+	d23e9e24-e734-44b3-95da-c75e9a876...	1	0	0
10	500-1000	cf036d38-2d85-4f4c-8fc9-dbc0af9155b2	1	0	0
11	1000+	f0dfd812-c54f-492f-8708-d6041f0b6a01	1	0	0
12	250-500	3bee9328-7f30-47cc-91ca-1f50afd93830	1	0	0
13	0-50	47e2aede-f9fd-45a6-9eec-29dc51d1df7a	1	0	0
14	250-500	6bd5b98b-9f96-4907-bef0-fb452e1cb351	1	0	0
15	0-50	8fb755f7-6230-4290-b288-2ded951550c6	1	0	0
16	100-250	a5a96472-b1d7-41d4-b84b-997cc8c70b...	1	0	0
17	250-500	b04fefd8-cc4e-45b2-9c6e-a79a689fa3e4	1	0	0
18	1000+	04b81f6b-1d28-4dda-86bf-394d5e0d8f6d	1	1	1
19	250-500	0839770b-8446-4b85-91cc-a0786665f9...	1	0	0
20	0-50	0c9bb9a9-6abd-4a32-ac26-b0d4d12d5...	1	0	0

Project 1 — E-commerce Drop-off Analysis

1st Step-

PRICE BAND FUNNEL SESSION SUMMARY

```sql
SELECT
    price_band,

    SUM(Has_View) AS View_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
        THEN 1 ELSE 0
    END) AS Cart_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
         AND Has_Purchase = 1
        THEN 1 ELSE 0
    END) AS Purchase_Sessions,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(SUM(Has_View), 0)
        AS DECIMAL(10,2)
    ) AS View_to_Cart_Conversion,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) * 100.0
        /
        NULLIF(
            SUM(CASE
                WHEN Has_View = 1
                 AND Has_Cart = 1
                THEN 1 ELSE 0
            END), 0
        )
        AS DECIMAL(10,2)
    ) AS Cart_to_Purchase_Conversion,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(SUM(Has_View), 0)
        AS DECIMAL(10,2)
    ) AS View_to_Purchase_Conversion

FROM dbo.price_session_funnel

GROUP BY price_band

ORDER BY View_to_Purchase_Conversion DESC;
```

	price_band	View_Sessions	Cart_Sessions	Purchase_Sessions	View_to_Cart_Conversion	Cart_to_Purchase_Conversion	View_to_Purchase_Conversion
1	100-250	46724	1920	1120	4.11	58.33	2.40
2	250-500	33504	1215	746	3.63	61.40	2.23
3	500-1000	23621	841	513	3.56	61.00	2.17
4	1000+	12805	419	237	3.27	56.56	1.85
5	50-100	27277	330	160	1.21	48.48	0.59
6	0-50	31283	481	177	1.54	36.80	0.57

Project 1 — E-commerce Drop-off Analysis

5th SQL Query-
TIME BASED SESSION FUNNEL

```sql
INSERT INTO dbo.hour_session_funnel
(
    event_hour,
    user_session,
    Has_View,
    Has_Cart,
    Has_Purchase
)
SELECT
    event_hour,
    user_session,

    MAX(CASE
        WHEN event_type = 'view' THEN 1
        ELSE 0
    END) AS Has_View,

    MAX(CASE
        WHEN event_type = 'cart' THEN 1
        ELSE 0
    END) AS Has_Cart,

    MAX(CASE
        WHEN event_type = 'purchase' THEN 1
        ELSE 0
    END) AS Has_Purchase

FROM dbo.ecommerce_events
WHERE event_hour IS NOT NULL
GROUP BY
    event_hour,
    user_session;

SELECT TOP 20 *          --- just checking
FROM dbo.hour_session_funnel
ORDER BY event_hour;
```

	event_hour	user_session	Has_View	Has_Cart	Has_Purchase
1	0	b62ab45e-b963-4f25-83e7-4999adea6cb3	1	0	0
2	0	18e33676-8d30-48e9-9142-d7cb1a4da8...	1	0	0
3	0	fa74f434-2a06-4b17-be7a-72c796087716	1	0	0
4	0	545c5b25-a2db-4b48-84e4-7e44e7b0cf82	1	0	0
5	0	3e063046-8877-4617-9241-c1d5fd912882	1	0	0
6	0	ea2b046d-b9ed-42bf-8f46-b4201c7e4040	1	0	0
7	0	74d40a28-41f9-4325-bbae-b179bd2c0a38	1	0	0
8	0	ab1c76e1-94e1-47a2-9084-04e8be98e7...	1	0	0
9	0	9333dfbd-b87a-4708-9857-6336556b0fcc	1	0	0
10	0	3ea7c620-a8d7-45c5-9ced-2e9874e2f549	1	0	0
11	0	1bb0afbf-abc2-469e-9fac-ef1976d669e1	1	0	0
12	0	4f2a5505-580d-4332-bfde-e76f27c60452	1	0	0
13	0	0e729c38-4aba-4961-8b6a-bfa37fcd555a	1	0	0
14	0	ba31f6de-daa2-def2-a02e-ecc10dff4ff9	1	0	0
15	0	7f0062d8-ead0-4e0a-96f6-43a0b79a2fc4	1	0	0
16	0	d8d1598c-7964-0480-a55d-efcf22220e4d	1	0	0
17	0	1035cee3-7238-4039-9c27-0e831fc0dc5c	1	0	0
18	0	d7c4761f-de75-454b-9164-177db5e53695	1	0	0
19	0	d88f3e93-ac85-4024-872c-be766a3c4199	1	0	0
20	0	85b6f12c-76f8-4816-9d84-549ad4741e58	1	0	0

Project 1 — E-commerce Drop-off Analysis

1st Step-
HOURLY SESSIONS FUNNEL SUMMARY

```sql
SELECT
    event_hour,

    SUM(Has_View) AS View_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
        THEN 1 ELSE 0
    END) AS Cart_Sessions,

    SUM(CASE
        WHEN Has_View = 1
         AND Has_Cart = 1
         AND Has_Purchase = 1
        THEN 1 ELSE 0
    END) AS Purchase_Sessions,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(SUM(Has_View), 0)
        AS DECIMAL(10,2)
    ) AS View_to_Cart_Conversion,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) * 100.0
        /
        NULLIF(
            SUM(CASE
                WHEN Has_View = 1
                 AND Has_Cart = 1
                THEN 1 ELSE 0
            END), 0
        )
        AS DECIMAL(10,2)
    ) AS Cart_to_Purchase_Conversion,

    CAST(
        SUM(CASE
            WHEN Has_View = 1
             AND Has_Cart = 1
             AND Has_Purchase = 1
            THEN 1 ELSE 0
        END) * 100.0
        / NULLIF(SUM(Has_View), 0)
        AS DECIMAL(10,2)
    ) AS View_to_Purchase_Conversion

FROM [dbo].[hour_session_funnel]

GROUP BY event_hour

ORDER BY event_hour;
```

	event_hour	View_Sessions	Cart_Sessions	Purchase_Sessions	View_to_Cart_Conversion	Cart_to_Purchase_Conversion	View_to_Purchase_Conversion
1	0	425	3	2	0.71	66.67	0.47
2	1	103	0	0	0.00	NULL	0.00
3	2	6097	165	92	2.71	55.76	1.51
4	3	12269	403	222	3.28	55.09	1.81
5	4	13782	536	274	3.89	51.12	1.99
6	5	15784	630	347	3.99	55.08	2.20
7	6	16349	733	412	4.48	56.21	2.52
8	7	17847	754	402	4.22	53.32	2.25
9	8	18537	789	425	4.26	53.87	2.29
10	9	17622	808	468	4.59	57.92	2.66
11	10	6917	307	159	4.44	51.79	2.30

Project 1 — E-commerce Drop-off Analysis

6th SQL Query-
BEHAVIOUR ANALYSIS

```sql
SELECT
    CASE
        WHEN View_Count = 1 THEN '1 View'
        WHEN View_Count BETWEEN 2 AND 3 THEN '2-3 Views'
        WHEN View_Count BETWEEN 4 AND 5 THEN '4-5 Views'
        ELSE '6+ Views'
    END AS View_Band,

    COUNT(*) AS Total_Sessions,

    SUM(Has_Purchase) AS Purchase_Sessions,

    CAST(
        SUM(Has_Purchase) * 100.0
        / NULLIF(COUNT(*), 0)
        AS DECIMAL(10,2)
    ) AS Purchase_Conversion

FROM
(
    SELECT
        user_session,

        COUNT(CASE
            WHEN event_type = 'view' THEN 1
        END) AS View_Count,

        MAX(CASE
            WHEN event_type = 'purchase' THEN 1
            ELSE 0
        END) AS Has_Purchase

    FROM dbo.ecommerce_events

    GROUP BY user_session
) AS Session_Behavior

GROUP BY
    CASE
        WHEN View_Count = 1 THEN '1 View'
        WHEN View_Count BETWEEN 2 AND 3 THEN '2-3 Views'
        WHEN View_Count BETWEEN 4 AND 5 THEN '4-5 Views'
        ELSE '6+ Views'
    END

ORDER BY
    MIN(View_Count);
```

	View_Band	Total_Sessions	Purchase_Sessions	Purchase_Conversion
1	6+ Views	24013	2198	9.15
2	1 View	44034	1462	3.32
3	2-3 Views	34712	3225	9.29
4	4-5 Views	14696	1371	9.33

Project 1 — E-commerce Drop-off Analysis

CART ENGAGEMENT V/S PURCHASE:-

```sql
SELECT
    CASE
        WHEN Has_Cart = 1 THEN 'Cart Added'
        ELSE 'No Cart'
    END AS Cart_Status,

    COUNT(*) AS Total_Sessions,

    SUM(Has_Purchase) AS Purchase_Sessions,

    CAST(
        SUM(Has_Purchase) * 100.0
        / NULLIF(COUNT(*), 0)
        AS DECIMAL(10,2)
    ) AS Purchase_Conversion

FROM dbo.session_funnel

GROUP BY
    Has_Cart

ORDER BY
    Has_Cart DESC;
```

	Cart_Status	Total_Sessions	Purchase_Sessions	Purchase_Conversion
1	Cart Added	5077	2882	56.77
2	No Cart	112378	5374	4.78

Project 1 — E-commerce Drop-off Analysis

Multiple Product Interaction:-

Do sessions interacting with multiple different products have higher purchase conversion?

```sql
SELECT
    CASE
        WHEN Product_Count = 1 THEN '1 Product'
        WHEN Product_Count BETWEEN 2 AND 3 THEN '2-3 Products'
        WHEN Product_Count BETWEEN 4 AND 5 THEN '4-5 Products'
        ELSE '6+ Products'
    END AS Product_Interaction_Band,

    COUNT(*) AS Total_Sessions,

    SUM(Has_Purchase) AS Purchase_Sessions,

    CAST(
        SUM(Has_Purchase) * 100.0
        / NULLIF(COUNT(*), 0)
        AS DECIMAL(10,2)
    ) AS Purchase_Conversion

FROM
(
    SELECT
        user_session,

        COUNT(DISTINCT product_id) AS Product_Count,

        MAX(
            CASE
                WHEN event_type = 'purchase' THEN 1
                ELSE 0
            END
        ) AS Has_Purchase

    FROM [dbo].[ecommerce_events]

    GROUP BY user_session
) AS Product_Behavior

GROUP BY
    CASE
        WHEN Product_Count = 1 THEN '1 Product'
        WHEN Product_Count BETWEEN 2 AND 3 THEN '2-3 Products'
        WHEN Product_Count BETWEEN 4 AND 5 THEN '4-5 Products'
        ELSE '6+ Products'
    END

ORDER BY
    MIN(Product_Count);
```

	Product_Interaction_Band	Total_Sessions	Purchase_Sessions	Purchase_Conversion
1	1 Product	59096	4339	7.34
2	2-3 Products	32665	2369	7.25
3	4-5 Products	11747	736	6.27
4	6+ Products	13947	812	5.82

Project 1 — E-commerce Drop-off Analysis

Purchase Conversion by Session Duration:-
Do longer sessions have different purchase conversion rates?

```sql
SELECT
    CASE
        WHEN Session_Duration_Minutes < 1 THEN 'Under 1 Min'
        WHEN Session_Duration_Minutes BETWEEN 1 AND 5 THEN '1-5 Min'
        WHEN Session_Duration_Minutes BETWEEN 6 AND 15 THEN '6-15 Min'
        WHEN Session_Duration_Minutes BETWEEN 16 AND 30 THEN '16-30 Min'
        ELSE '30+ Min'
    END AS Duration_Band,

    COUNT(*) AS Total_Sessions,

    SUM(Has_Purchase) AS Purchase_Sessions,

    CAST(
        SUM(Has_Purchase) * 100.0
        / NULLIF(COUNT(*), 0)
        AS DECIMAL(10,2)
    ) AS Purchase_Conversion

FROM
(
    SELECT
        user_session,

        DATEDIFF(
            MINUTE,
            MIN(event_time),
            MAX(event_time)
        ) AS Session_Duration_Minutes,

        MAX(
            CASE
                WHEN event_type = 'purchase' THEN 1
                ELSE 0
            END
        ) AS Has_Purchase

    FROM [dbo].[ecommerce_events]

    GROUP BY user_session
) AS Session_Duration
```

```sql
) AS Session_Duration

GROUP BY
    CASE
        WHEN Session_Duration_Minutes < 1 THEN 'Under 1 Min'
        WHEN Session_Duration_Minutes BETWEEN 1 AND 5 THEN '1-5 Min'
        WHEN Session_Duration_Minutes BETWEEN 6 AND 15 THEN '6-15 Min'
        WHEN Session_Duration_Minutes BETWEEN 16 AND 30 THEN '16-30 Min'
        ELSE '30+ Min'
    END

ORDER BY
    MIN(
        CASE
            WHEN Session_Duration_Minutes < 1 THEN 0
            WHEN Session_Duration_Minutes BETWEEN 1 AND 5 THEN 1
            WHEN Session_Duration_Minutes BETWEEN 6 AND 15 THEN 6
            WHEN Session_Duration_Minutes BETWEEN 16 AND 30 THEN 16
            ELSE 31
        END
    );

SELECT
    COUNT(*) AS Total_Rows,
    COUNT(DISTINCT user_session) AS Unique_Sessions,
    SUM(Has_Purchase) AS Purchase_Flag_Total,
    COUNT(DISTINCT CASE
        WHEN Has_Purchase = 1 THEN user_session
    END) AS Purchase_Sessions
FROM dbo.session_funnel;
```

	Duration_Band	Total_Sessions	Purchase_Sessions	Purchase_Conversion
1	Under 1 Min	50111	247	0.49
2	1-5 Min	44835	5015	11.19
3	6-15 Min	15168	2101	13.85
4	16-30 Min	4308	549	12.74
5	30+ Min	3033	344	11.34

	Total_Rows	Unique_Sessions	Purchase_Flag_Total	Purchase_Sessions
1	117455	117455	8256	8256